



## CONTACT

EMAIL  
sangeethsasikumar47@gmail.com

PORTFOLIO  
sangeeth2008-github-io.vercel.app

GITHUB  
github.com/sangeeth2008

LINKEDIN  
linkedin.com/in/sangeeth-sasikumar-k-s

LOCATION  
Kerala, India

## TECHNICAL SKILLS

### LANGUAGES

C / C++ Python

### PLATFORMS

ESP32 ESP32-CAM Arduino  
FreeRTOS PlatformIO

### ROBOTICS

Differential drive FSM design  
PID control Motor drivers  
Obstacle avoidance

### IOT & PROTOCOLS

MQTT Blynk REST WebSockets  
MJPEG I2C SPI UART PWM

### EDGE AI & VISION

Edge Impulse TensorFlow Lite Micro  
INT8 quantization OpenCV

## EDUCATION

**B.Tech, Mechatronics Engineering**  
Jyothi Engineering College (Autonomous),  
Kerala  
Currently pursuing

*"Good Code. Cool Robots.  
Better Me."*

# SANGEETH SASIKUMAR K S

MECHATRONICS ENGINEER · ROBOTICS · EMBEDDED SYSTEMS · EDGE AI

## PROFILE

Mechatronics undergraduate who learns by building. I design ESP32-based robots and IoT systems that sense, decide and act on their own: non-blocking C/C++ firmware, hardware failsafes, and TinyML vision that runs on the device instead of the cloud. My portfolio spans security and firefighting robots, AI waste sorting, crop monitoring and gas-safety nodes.

## SELECTED PROJECTS

### Smart Security Patrol Robot V1 **FLAGSHIP**

ESP32 dual-core · ESP32-CAM · TB6612FNG · HC-SR04 · MPU6050 · PIR

- Autonomous differential-drive rover run by a non-blocking finite state machine.
- Multi-zone ultrasonic obstacle clearance, PIR intrusion trigger and MPU6050 tilt-protection cutoff.
- Onboard ESP32-CAM serves live MJPEG video over Wi-Fi.

### AI Waste Segregation Machine **TINYML**

Edge Impulse · quantized CNN · ESP32 · inductive sensor · servo diverters

- Camera vision with an on-device neural network classifies waste as biodegradable, plastic or metal.
- Drives multi-chute sorting gates through servo diverters.

### Smart Agriculture & AI Crop Monitoring **EDGE AI + IOT**

ESP32-CAM · Edge Impulse INT8 · Blynk · capacitive soil probe · DHT22

- Real-time plant disease detection on-device, combined with soil and microclimate telemetry to the Blynk cloud.

### Smart Waste Bin (Wet / Dry) **ESP32**

ESP32 · capacitive moisture sensor · DHT22 · servo · SSD1306 OLED · SoftAP

- Sorts waste with a servo flap, raises a DHT22 fire alarm, shows status on an OLED.
- Hosts its own Wi-Fi dashboard (SoftAP), so it needs no router or internet.

### MORE BUILDS

**Fire Fighting Robot** – flame-sensor array, obstacle avoidance, submersible pump with dual-servo nozzle aiming.

**Smart LPG Safety System** – MQ-2 gas detection, servo valve cutoff, piezo siren, MQTT push alerts.

**Smart Classroom System** – RFID RC522 authentication, climate-driven relay ventilation, I2C LCD readout.

**Personal AI "MARK"** – Python assistant for lab telemetry diagnostics, voice interaction and test automation.

## CURRENTLY BUILDING

**Rover V2** – ESP32 dual-core controller, dual HC-SR04 ultrasonic zones, ESP32-CAM vision, 3S LiPo power, differential-drive kinematics with PID speed control.

## HOW I BUILD

- Deterministic firmware.** Non-blocking state machines, interrupts and FreeRTOS tasks instead of delay-driven code.
- Failsafes first.** Watchdogs, brownout recovery and tilt or gas cutoffs designed in from day one.
- Edge over cloud.** Quantized models and local dashboards so systems keep working offline.